Chapter 7 — All 9 Lecture Examples

black = write it down · **red = the answer** · *teal = where it came from / my note* · grey = optional

Every example is solved the professor's way. Where he skipped algebra, the missing lines are filled in and marked. Each one opens with **Given → Want → Route** so the flow is visible before the algebra starts, and every imported equation carries the slide it comes from.

Ex 1	Faraday Rotation — find z where the wave is y-polarized	Lec 2 s19, Lec 3 s2–4
Ex 2	Faraday Rotation applied — the angle θ(z) **[HW 1]**	Lec 3 s5
Ex 3	Total Internal Reflection — the critical angle	Lec 4 s7
Ex 4	Optical Fiber — largest guided ray angle	Lec 4 s11–12
Ex 5	Brewster's angle **[HW 2]**	Lec 4 s13
Ex 6	X-ray — almost total external reflection, $R \propto q^{-4}$	Lec 6 s4–5
Ex 7	Plasma — ε(ω) and the plasma frequency	Lec 6 s6–7
Ex 8	Reflection from a good conductor at normal incidence	Lec 7 s9
Ex 9	Gaussian modulated oscillation — a spreading pulse	Lec 9 s2–5

Exam scope. *Chapter 7 runs from Lec 1 to* ***Lec 9 slide 9****. Lec 9 slides 10–17 (waveguides, cavities, 2D Laplace) are Jackson* ***Chapter 8*** *and are not in Chapter 7.*

Example 1 — Faraday Rotation

Lec 2 s19, Lec 3 s2–4

> For propagation along the z-axis, a medium supports left circular polarization with index n_L and right circular polarization with index n_R. A plane wave in this medium has $\mathbf{E}(z = 0, t) = \hat{\mathbf{x}} E \exp(-i\omega t)$. Find the values of z where the wave is **linearly polarized along the y-axis**.

Flow: split the wave into LCP + RCP → fix both from $z = 0$ → let each run at its own speed → set $E_x = 0$.

These are his slide lines, in his order. One correction is marked in Step 4.

Step 1 — Write it as LCP + RCP

The medium gives the two circular polarizations different indices, so each needs its own k.

$$\vec{E}(x,t) = \vec{E}_L e^{i(k_L z - \omega t)} + \vec{E}_R e^{i(k_R z - \omega t)}$$ Lec 3, s2

$$\vec{E}_L = \varepsilon_L \hat{e}_+, \qquad \vec{E}_R = \varepsilon_R \hat{e}_-$$

$$\hat{e}_+ = \frac{\sqrt{2}}{2}(\hat{\mathbf{x}} + i\hat{\mathbf{y}}), \qquad \hat{e}_- = \frac{\sqrt{2}}{2}(\hat{\mathbf{x}} - i\hat{\mathbf{y}})$$ Lec 2, s15

Step 2 — Put z = 0 into the Step 1 equation

$z = 0$ is the **only** place the problem tells us the field, so it is the only place we can pin down ε_L and ε_R. Setting $z = 0$ above, both exponentials become $e^{-i\omega t}$:

$$\vec{E}(0,t) = \vec{E}_L e^{-i\omega t} + \vec{E}_R e^{-i\omega t}$$

and the problem states what that equals:

$$\vec{E}(0,t) = \hat{\mathbf{x}} E e^{-i\omega t}$$ *given in the question*

Set the two equal:

$$\vec{E}_L e^{-i\omega t} + \vec{E}_R e^{-i\omega t} = \hat{\mathbf{x}} E e^{-i\omega t}$$ Lec 3, s2

$e^{-i\omega t}$ is never zero, so cancel it from all three terms (this is the strike-through on his slide), then put in $\vec{E}_L = \varepsilon_L \hat{e}_+$ and $\vec{E}_R = \varepsilon_R \hat{e}_-$:

$$\varepsilon_L \hat{e}_+ + \varepsilon_R \hat{e}_- = \hat{\mathbf{x}} E$$

Step 3 — Match the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ parts

$$\varepsilon_L \frac{\sqrt{2}}{2}(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) + \varepsilon_R \frac{\sqrt{2}}{2}(\hat{\mathbf{x}} - i\hat{\mathbf{y}}) = \hat{\mathbf{x}}E$$ Lec 3, s3

Multiply out and collect the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ parts:

direction	left side	right side
$\hat{\mathbf{x}}$	$\frac{\sqrt{2}}{2}(\varepsilon_L + \varepsilon_R)$	E
$\hat{\mathbf{y}}$	$i\frac{\sqrt{2}}{2}(\varepsilon_L - \varepsilon_R)$	0

*$\hat{x}$ and $\hat{y}$ are independent directions — no amount of $\hat{x}$ can make a $\hat{y}$. So the two rows must hold **separately**. Same as resolving forces: if the resultant points purely along x, the y-components must sum to zero.*

The $\hat{\mathbf{y}}$ row: i and $\frac{\sqrt{2}}{2}$ are not zero, so the bracket must be:

$$i\frac{\sqrt{2}}{2}(\varepsilon_L - \varepsilon_R) = 0 \quad \Rightarrow \quad \varepsilon_L = \varepsilon_R$$

The $\hat{\mathbf{x}}$ row, with ε_R replaced by ε_L:

$$\frac{\sqrt{2}}{2}(\varepsilon_L + \varepsilon_L) = E \Rightarrow \frac{\sqrt{2}}{2}(2\varepsilon_L) = E \Rightarrow \sqrt{2}\,\varepsilon_L = E$$

$$\varepsilon_L = \frac{E}{\sqrt{2}} = \frac{E}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{E\sqrt{2}}{2}$$ *same number, no root underneath*

$$\varepsilon_L = \varepsilon_R = \frac{\sqrt{2}}{2}E$$

What this means. *Equal amounts of left- and right-circular. That is what a linear polarization is: two counter-rotating circles of the same size add up to a straight line. Unequal amounts would give an ellipse.*

Step 4 — Put them back, take the x-part

$$\vec{E}(x,t) = \frac{\sqrt{2}}{2} E \hat{e}_+ e^{i(k_L z-\omega t)} + \frac{\sqrt{2}}{2} E \hat{e}_- e^{i(k_R z-\omega t)}$$ Lec 3, s3

Now put $\hat{e}_+$ and $\hat{e}_-$ in **full**, from Step 1:

$$\vec{E} = \frac{\sqrt{2}}{2} E \cdot \frac{\sqrt{2}}{2} (\hat{\mathbf{x}}+i\hat{\mathbf{y}}) e^{i(k_L z-\omega t)} + \frac{\sqrt{2}}{2} E \cdot \frac{\sqrt{2}}{2} (\hat{\mathbf{x}}-i\hat{\mathbf{y}}) e^{i(k_R z-\omega t)}$$

The number in front of each term is $\frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2}$:

$$\sqrt{2} \times \sqrt{2} = 2, \quad 2 \times 2 = 4, \quad \Rightarrow \quad \frac{2}{4} = \frac{1}{2}$$

$$\vec{E} = \frac{1}{2} E (\hat{\mathbf{x}} + i\hat{\mathbf{y}}) e^{i(k_L z-\omega t)} + \frac{1}{2} E (\hat{\mathbf{x}} - i\hat{\mathbf{y}}) e^{i(k_R z-\omega t)}$$

Collect $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, same as Step 3:

direction	what multiplies it
$\hat{\mathbf{x}}$	$\frac{1}{2} E e^{i(k_L z-\omega t)} + \frac{1}{2} E e^{i(k_R z-\omega t)}$
$\hat{\mathbf{y}}$	$\frac{1}{2} iE e^{i(k_L z-\omega t)} - \frac{1}{2} iE e^{i(k_R z-\omega t)}$

The top row is E_x:

$$E_x(\vec{x},t) = \frac{1}{2} E \left(e^{i(k_L z-\omega t)} + e^{i(k_R z-\omega t)} \right)$$

Correction. *The slide writes ½ i E here. Look at the table above: the* ***i*** *sits in the* $\hat{\mathbf{y}}$ *row, and that row has a* ***minus*** *between the exponentials. He took the i from* E_y *but kept the plus of* E_x*. It does not change the final answer (a constant in front cannot move a zero), but* E_x *has no i.*

Step 5 — Along y means $E_x = 0$

$e^{i(k_L z - \omega t)} = -e^{i(k_R z - \omega t)}$ Lec 3, s4

Two numbers of the same size are opposite when their phases differ by an **odd** multiple of π:

$(k_L z - \omega t) - (k_R z - \omega t) = (2m+1)\pi$ *m is an integer*

$\Rightarrow k_L z - k_R z = (2m+1)\pi$ *ωt cancels — the answer does not depend on time*

Step 6 — Put in $k = \omega n/c$

$k = \frac{\omega n}{c}$ Lec 2, s18

ANSWER

$$z = \frac{(2m+1)\pi}{k_L - k_R} = \frac{(2m+1)\pi c}{\omega(n_L - n_R)}$$

m = 0, 1, 2, … — a series of planes, evenly spaced.

OPTIONAL **Faster route.** Example 2 gives $\theta(z) = -(k_L - k_R)z/2$. Along $\hat{y}$ means $\theta = \pm 90°$, which gives the same z in one line. If both come up in the exam, do Example 2 first.

Example 2 — Application of Faraday Rotation

Lec 3 s5

> With $\tan\theta = E_y/E_x$, find the rotation angle **θ(z)** of the polarization direction. **[He wrote "HW 1" here and left it unsolved.]**

Given: the same field as Example 1. **Want:** θ(z), the angle the polarization line makes with $\hat{\mathbf{x}}$.

Route: take E_x and E_y from Example 1 → pull out the common phase → the ratio becomes a single tangent.

Step 1 — Both components, from Example 1

E_x is Step 3 of Example 1. For E_y, note $\hat{e}_+$ carries $+i/\sqrt{2}$ of $\hat{\mathbf{y}}$ and $\hat{e}_-$ carries $-i/\sqrt{2}$:

$$E_x = \frac{E}{2}\left(e^{ik_L z} + e^{ik_R z}\right)e^{-i\omega t}$$

$$E_y = \frac{iE}{2}\left(e^{ik_L z} - e^{ik_R z}\right)e^{-i\omega t}$$

Step 2 — Pull out the common phase

Split each wavenumber into **average ± half-difference**:

$$\bar{k} = \frac{k_L + k_R}{2}, \qquad \Delta = \frac{(k_L - k_R)z}{2}$$

*Δ is an **angle**, not a wavenumber — the z and the ½ are inside it*

$$k_L z = \bar{k}z + \Delta, \qquad k_R z = \bar{k}z - \Delta$$

Same as writing 7 and 3 as 5 + 2 and 5 − 2: mean, plus and minus half the gap. Check: $\bar{k}z + \Delta = (k_L+k_R)z/2 + (k_L-k_R)z/2 = k_L z$ ✓

Substitute, then split and factor the exponentials:

$$e^{ik_L z} + e^{ik_R z} = e^{i(\bar{k}z + \Delta)} + e^{i(\bar{k}z - \Delta)}$$

$$= e^{i\bar{k}z}e^{i\Delta} + e^{i\bar{k}z}e^{-i\Delta} = e^{i\bar{k}z}\left(e^{i\Delta} + e^{-i\Delta}\right)$$

The bracket is now **Euler's formula**. Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$:

$e^{i\theta} + e^{-i\theta} = 2\cos\theta$ *the sines cancel*

$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$ *the cosines cancel*

So the two combinations are:

$e^{ik_L z} + e^{ik_R z} = 2\,e^{i\bar{k}z}\cos\Delta$

$e^{ik_L z} - e^{ik_R z} = 2i\,e^{i\bar{k}z}\sin\Delta$

Why this helps. *Before, E_x and E_y were sums of two exponentials that would not combine. Now both carry the* ***same*** *factor $e^{i\bar{k}z}$ and differ only by $\cos\Delta$ against $\sin\Delta$ — so dividing them gives a tangent, which is exactly what $\tan\theta = E_y/E_x$ needs.*

Step 3 — Substitute back

$$E_x = E\, e^{i(\bar{k}z-\omega t)} \cos\Delta$$

$$E_y = \frac{iE}{2}\,(2i\, e^{i\bar{k}z} \sin\Delta)\, e^{-i\omega t} = -E\, e^{i(\bar{k}z-\omega t)} \sin\Delta \qquad i \times i = -1$$

*Both components now carry the **same** phase factor $e^{i(\bar{k}z-\omega t)}$. That is what makes the wave **linearly** polarized at every z — only the direction of the line turns.*

Step 4 — Form the ratio

$$\tan\theta = \frac{E_y}{E_x} = -\frac{\sin\Delta}{\cos\Delta} = -\tan\Delta \quad \Rightarrow \quad \theta = -\Delta$$

Finally put Δ back, with $k = \omega n/c$:

ANSWER

$$\theta(z) = -\frac{(k_L - k_R)z}{2} = \frac{\omega}{2c}(n_R - n_L)z$$

*The plane of polarization turns **steadily** with z, at the constant rate $d\theta/dz = (\omega/2c)(n_R - n_L)$.*

OPTIONAL **Consistency with Example 1.** The wave is along $\hat{y}$ when $\theta = \pm\pi/2$, i.e. $\Delta = (2m+1)\pi/2$. That gives $z = (2m+1)\pi/(k_L - k_R)$ — exactly the answer to Example 1. ✓

Example 3 — Total Internal Reflection (Lec 4 s7)

> When n > n′, the refraction angle r exceeds the incidence angle i. **What happens if sin r > 1?**

Given: wave going from dense (n) to less dense (n′). **Want:** when the transmitted wave stops existing.
Route: Snell → sin r can exceed 1 → cos r turns imaginary → critical angle.

Step 1 — Snell's law

$n \sin i = n' \sin r$ *from phase matching at the surface* (Lec 3, s12)

$$\sin r = \frac{n}{n'} \sin i$$

Step 2 — If n > n′, this can exceed 1

$n/n' > 1$, so for large enough i the right-hand side passes 1 — and no *real* angle r has sin r > 1.

$\cos r = \sqrt{1 - \sin^2 r}$ ⇒ **cos r is imaginary** when sin r > 1

*An imaginary cos r means the transmitted wave no longer propagates — it becomes an **evanescent** wave that decays away from the surface.* (Lec 4, s8)

Step 3 — The boundary case sin r = 1

The changeover is exactly where sin r = 1, i.e. r = 90°:

$$\frac{n}{n'} \sin i_0 = 1$$

ANSWER

$$i_0 = \arcsin \frac{n'}{n} \quad — \quad \text{if } i > i_0, \textbf{ total internal reflection}$$

Glass to air (n = 1.5, n′ = 1): $i_0 = 41.8°$.

Example 4 — Optical Fiber

Lec 4 s11–12

> A fiber has core index n_1 and cladding index n_0, with $n_0 < n_1$ (difference only ~1%). Find the largest angle $\boldsymbol{\theta_{max}}$, measured from the fiber **axis**, for which a ray stays trapped in the core.

Given: n_1 core, n_0 cladding. **Want:** θ_{max} from the axis.
Route: trapping = TIR at the core wall → but θ is measured from the axis, not the wall normal → convert.

Step 1 — Trapping means TIR at the wall

The ray must be totally internally reflected every time it hits the core–cladding boundary. From Example 3, with the core as the dense medium:

$$i > i_0, \quad \sin i_0 = \frac{n_0}{n_1}$$ *Example 3, with $n \to n_1$, $n' \to n_0$*

Step 2 — Convert the wall angle to the axis angle

i is measured from the **wall normal**, which points across the fiber. θ is measured from the **axis**, which runs along it. The two directions are perpendicular, so

$$i = 90° - \theta \quad \Rightarrow \quad \sin i = \cos\theta$$ *this is the only geometry step — read it off the figure*

Step 3 — Rewrite the condition

$$\cos\theta > \frac{n_0}{n_1}$$

cos is decreasing, so a *larger* cos means a *smaller* θ. The limit is

ANSWER

$$\theta_{max} = \cos^{-1}\frac{n_0}{n_1}$$

$n_0 = 1.45$, $n_1 = 1.465$ gives $\theta_{max} \approx 8.2°$ — only rays almost parallel to the axis are guided.

Example 5 — Brewster's angle

Lec 4 s13

> Assume $\mu = \mu'$. For a **p-polarized** wave, find the incidence angle at which the reflected wave vanishes. **[He wrote "HW 2" before the last algebra step.]**

Given: p-polarization, $\mu = \mu'$. **Want:** i_B with $E_o'' = 0$.

Route: p-polarized reflection coefficient → set its numerator to zero → bring in $Z = \mu c/n$ and Snell → solve for tan i.

Step 1 — The p-polarized reflection coefficient

$$\frac{E_o''}{E_o} = \frac{Z\cos i - Z'\cos r}{Z\cos i + Z'\cos r}$$

Lec 4, s2–3

Step 2 — No reflection means the numerator vanishes

$$E_o'' = 0 \;\Rightarrow\; Z\cos i = Z'\cos r$$

Step 3 — Turn the impedances into indices

$$Z = \sqrt{\mu/\varepsilon} = \frac{\mu c}{n}$$

Lec 2, s18

With $\mu = \mu'$ the μc cancels in the ratio, leaving

$$\frac{Z}{Z'} = \frac{n'}{n} \;\Rightarrow\; n'\cos i = n\cos r \quad (1)$$

Step 4 — Bring in Snell's law

$$n\sin i = n'\sin r \quad (2)$$

Lec 3, s12

Two equations, two unknowns (i and r). Eliminate r — this is the part left as HW 2.

Step 5 — Eliminate r

From ①: $\cos r = (n'/n)\cos i$. From ②: $\sin r = (n/n')\sin i$. Put both into $\sin^2 r + \cos^2 r = 1$:

$$\frac{n'^2}{n^2}\cos^2 i + \frac{n^2}{n'^2}\sin^2 i = 1$$

Multiply through by $n^2 n'^2$:

$$n'^4\cos^2 i + n^4\sin^2 i = n^2 n'^2$$

Divide by $\cos^2 i$ and use $1/\cos^2 i = 1 + \tan^2 i$:

$$n'^4 + n^4\tan^2 i = n^2 n'^2(1 + \tan^2 i)$$

Collect the $\tan^2 i$ terms:

$$\tan^2 i\,(n^4 - n^2 n'^2) = n^2 n'^2 - n'^4$$

$$\tan^2 i\, n^2(n^2 - n'^2) = n'^2(n^2 - n'^2) \quad \text{the bracket cancels}$$

$$\tan^2 i = \frac{n'^2}{n^2}$$

ANSWER

$$i_B = \arctan\frac{n'}{n}$$

Air to glass ($n = 1$, $n' = 1.5$): $i_B = 56.3°$. At this angle the reflected light is **purely s-polarized** — which is what a polarizing filter exploits.

OPTIONAL Substituting i_B back gives $i_B + r = 90°$ exactly: at Brewster's angle the reflected and transmitted rays are perpendicular.

Example 6 — X-ray: almost total external reflection

Lec 6 s4–5

At X-ray frequencies the index of essentially all matter is slightly **less** than 1: $n \approx 1 - \delta$ with $\delta \sim 10^{-5}$. A plane wave in vacuum with wave vector $\mathbf{k}_I$ reflects from a non-magnetic sample; let $\mathbf{q} = \mathbf{k}_R - \mathbf{k}_I$. Show that R_s and R_p both vary as $\mathbf{q}^{-4}$ when the grazing angle is small, but not small enough for total external reflection.

Given: $n = 1 - \delta$, $\delta \ll 1$; grazing angles α_1 (in), α_2 (out). **Want:** $R \propto q^{-4}$.

Route: Snell in grazing angles → small-angle expand to get $\alpha_1^2 - \alpha_2^2 = 2\delta$ → Fresnel → swap α_1 for q.

Why grazing angles. *Here the angles are measured from the* ***surface*** *(α), not from the normal (θ), because X-rays only reflect appreciably when they skim the sample. $\alpha = 90° - \theta$, so $\sin\theta$ becomes $\cos\alpha$.*

Step 1 — Snell, rewritten in grazing angles

$$\sin\theta_1 = n\sin\theta_2 \quad \Rightarrow \quad \cos\alpha_1 = n\cos\alpha_2$$ Lec 3, s12

Step 2 — Expand for small angles

Both grazing angles are tiny, so use $\cos\alpha \approx 1 - \alpha^2/2$:

$$1 - \frac{\alpha_1^2}{2} = n - n\,\frac{\alpha_2^2}{2}$$

Put $n = 1 - \delta$. In the last term $n \approx 1$, since $\delta\alpha^2$ is negligible against everything else:

$$1 - \frac{\alpha_1^2}{2} = 1 - \delta - \frac{\alpha_2^2}{2}$$

$$\alpha_1^2 - \alpha_2^2 = 2\delta \qquad \text{the key relation — used in Step 5}$$

Total external reflection happens when $\alpha_1^2 < 2\delta$ (then α_2 has no real solution). The problem says we are above that, so $\alpha_1 > \sqrt{2\delta}$.

Step 3 — Fresnel for s-polarization, in grazing angles

$$r_s = \frac{\cos\theta_1 - n\cos\theta_2}{\cos\theta_1 + n\cos\theta_2} = \frac{\sin\alpha_1 - n\sin\alpha_2}{\sin\alpha_1 + n\sin\alpha_2}$$ Lec 4, s4

Step 4 — Small angles again, and n ≈ 1

$$R_s = |r_s|^2 = \left(\frac{\alpha_1 - \alpha_2}{\alpha_1 + \alpha_2}\right)^2 \quad \sin\alpha \approx \alpha$$

Step 5 — Bring in Step 2

Multiply numerator and denominator by $(\alpha_1 + \alpha_2)$, so the numerator becomes the difference of squares we already know:

$$R_s = \left(\frac{\alpha_1^2 - \alpha_2^2}{(\alpha_1 + \alpha_2)^2}\right)^2 = \left(\frac{2\delta}{(\alpha_1 + \alpha_2)^2}\right)^2$$

Well above the critical angle $\alpha_2 \approx \alpha_1$, so $(\alpha_1+\alpha_2)^2 \approx 4\alpha_1^2$:

$$R_s \approx \left(\frac{\delta}{2\alpha_1^2}\right)^2 \propto \alpha_1^{-4}$$

Step 6 — Replace α_1 by q

q is the change in wave vector on reflection. Incident and reflected have the same length k and are mirror images in the surface, so the difference is perpendicular to the surface with magnitude

$$q = 2k\sin\alpha_1 \approx 2k\alpha_1 \quad \Rightarrow \quad \alpha_1 = \frac{q}{2k}$$

ANSWER

$$R_s \approx \left(\frac{\delta}{2\alpha_1^2}\right)^2 = \left(\frac{2k^2\delta}{q^2}\right)^2 \propto q^{-4}$$

R_p goes the same way: at grazing incidence the s and p formulas coincide.

Example 7 — Plasma Lec 6 s6–7

> A plasma is a gas of **free** electrons: there is no restoring force ($\omega_0 = 0$) and no damping ($\gamma = 0$). Find $\varepsilon(\omega)$, define the plasma frequency, and say which waves can pass through.

Given: free electrons, density N, $\omega_0 = 0$, $\gamma = 0$. **Want:** $\varepsilon(\omega)/\varepsilon_0$ and the cut-off.
Route: take the bound-charge result from Lec 5 and switch off the two terms that free electrons do not have.

Step 1 — Start from the harmonically bound charge model

This is the general dielectric constant already derived in lecture — nothing new is introduced here, we only specialise it:

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = 1 + \frac{Ne^2}{\varepsilon_0 m} \sum_j f_j \left(\omega_j^2 - \omega^2 - i\omega\gamma_j \right)^{-1}$$ Lec 5, s7

Step 2 — Switch off binding and damping

Free electrons: one species, $\omega_0 = 0$ (nothing pulls them back) and $\gamma = 0$ (no collisions). The bracket collapses to $(-\omega^2)$:

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \frac{Ne^2}{\varepsilon_0 m} \frac{1}{-\omega^2} = 1 - \frac{Ne^2}{\varepsilon_0 m \omega^2}$$

Step 3 — Name the constant

Everything in front of $1/\omega^2$ is fixed by the material, so give it a name. It has units of frequency squared:

$$\omega_p^2 = \frac{Ne^2}{\varepsilon_0 m}$$ *the plasma frequency*

ANSWER

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2}, \qquad \omega_p = \sqrt{Ne^2/\varepsilon_0 m}$$

Step 4 — Read off the physics

If $\omega < \omega_p$ then $\varepsilon/\varepsilon_0 < 0$, so $n = \sqrt{\varepsilon/\varepsilon_0}$ is **imaginary** — the wave cannot propagate and is reflected. Above ω_p it passes through.

> OPTIONAL **His two numbers.** Laboratory plasma, $N \sim 10^{18}$ m^{-3} → $\omega_p \approx 6\times10^{10}$ rad/s — microwaves. Ionosphere, $N \sim 10^{12}$ m^{-3} → $\omega_p \approx 6\times10^{7}$ rad/s — radio. That is why AM radio bounces off the ionosphere and reaches beyond the horizon, while higher frequencies go straight through to satellites.
> (The slide labels these "Hz"; they are ω_p in rad/s. As an ordinary frequency, divide by 2π.)

Example 8 — Reflection from a good conductor, normal incidence

Lec 7 s9

> What is the reflection coefficient of a **good conductor** at normal incidence ($\omega \ll \gamma_0$)?

Given: insulator n_1 above, conductor $n_2 = n_r + i n_i$ below, normal incidence. **Want:** R.
Route: Fresnel at $i = 0$ → take the modulus squared → use $n_r = n_i$ for a good conductor → expand for $n_r \gg n_1$.

Step 1 — Fresnel at normal incidence

At $i = r = 0$ both cosines are 1 and the s and p formulas become the same:

$$R = \left| \frac{n_r + i n_i - n_1}{n_r + i n_i + n_1} \right|^2$$ Lec 4, s2–4

Step 2 — Write out the modulus

$|a + ib|^2 = a^2 + b^2$, applied top and bottom:

$$R = \frac{(n_r - n_1)^2 + n_i^2}{(n_r + n_1)^2 + n_i^2}$$

Step 3 — Use the good-conductor property $n_i = n_r$

His slide gives, for a good conductor ($\sigma \gg \varepsilon_r \omega$):

$$n_r = n_i = c\sqrt{\frac{\mu\sigma}{2\omega}}$$ *both parts of n are equal in a good conductor* Lec 7, s3

The medium is non-magnetic ($\mu = \mu_0$), and $c^2 = 1/\mu_0\varepsilon_0$, so $c^2\mu_0 = 1/\varepsilon_0$. Pulling the c inside the root:

$$n_r = \sqrt{c^2\mu_0\sigma / 2\omega} = \sqrt{\frac{\sigma_2}{2\varepsilon_1\omega}}$$ *same quantity, written without c and μ*

Substituting $n_i = n_r$ and expanding the squares:

$$R = \frac{2n_r^2 - 2n_r n_1 + n_1^2}{2n_r^2 + 2n_r n_1 + n_1^2} = 1 - \frac{4n_r n_1}{2n_r^2 + 2n_r n_1 + n_1^2}$$

The second form comes from writing numerator = denominator − $4n_r n_1$. It is the useful form because R is close to 1, and this isolates the small part.

Step 4 — A conductor has $n_r \gg n_1$

Of the three terms downstairs, $2n_r^2$ is far the largest:

$$R \approx 1 - \frac{4n_r n_1}{2n_r^2} = 1 - \frac{2n_1}{n_r}$$

Step 5 — Put n_r in terms of σ and ω

n_1 is the index of the insulator above, so it carries ε_1:

$$n_1 = c\sqrt{\mu_1\varepsilon_1}, \qquad n_r = c\sqrt{\frac{\mu_2\sigma_2}{2\omega}}$$ Lec 2 s18 · Lec 7 s3

Divide. The **c cancels** top and bottom:

$$\frac{2n_1}{n_r} = \frac{2c\sqrt{\mu_1\varepsilon_1}}{c\sqrt{\frac{\mu_2\sigma_2}{2\omega}}} = 2\sqrt{\mu_1\varepsilon_1 \cdot 2\omega / \mu_2\sigma_2}$$

Both media are non-magnetic, $\mu_1 = \mu_2$, so they cancel too:

$$= 2\sqrt{\frac{2\varepsilon_1\omega}{\sigma_2}}$$

Finally take the 2 inside the root. Since $2 = \sqrt{4}$, $2\sqrt{X} = \sqrt{4}\,\sqrt{X} = \sqrt{4X}$:

$$= \sqrt{4} \times 2\varepsilon_1\omega/\sigma_2 = \sqrt{\frac{8\varepsilon_1\omega}{\sigma_2}}$$ $4 \times 2 = 8$ — that is where the 8 comes from

ε_1 is there because it arrived with n_1, not from ε_0. The result holds for **any** insulator above, not just vacuum — checked against the exact Fresnel formula with glass above ($n_1 = 1.5$): 0.99958811 against 0.99958803.

ANSWER

$$R \approx 1 - \frac{2n_1}{n_r} = 1 - \sqrt{\frac{8\varepsilon_1\omega}{\sigma_2}}$$

The **Hagen–Rubens relation** Lec 7, s10. $R \to 1$ as $\omega \to 0$ or $\sigma \to \infty$ — a perfect conductor is a perfect mirror.

OPTIONAL Copper ($\sigma = 5.9\times10^7$ S/m) at 10 GHz: $R = 0.99972$ — only 0.03% of the power is absorbed per bounce.

Example 9 — Gaussian modulated oscillation

Lec 9 s2–5

> A pulse starts as $u(x,0) = e^{-x^2/2L^2}\cos k_0 x$ with $\partial u/\partial t(x,0) = 0$, in a medium with dispersion relation $\omega(k) = v(1 + a^2k^2/2)$. Find $u(x,t)$.

Given: the initial shape, zero initial velocity, and $\omega(k)$. **Want:** $u(x,t)$.

Route: 7.91 gives $A(k)$ → the integral is Gaussian, so it can be done exactly → put $A(k)$ into 7.90 → read off speed and spreading.

Step 1 — A(k) from the initial conditions

Because $\partial u/\partial t(x,0) = 0$, the second term of Eq. 7.91 drops out and only the initial shape is left:

$$A(k) = \frac{1}{\sqrt{2\pi}} \int dx\; e^{-ikx}\, u(x,0)$$ *Jackson 7.91 with $\partial u/\partial t(x,0) = 0$* Lec 8, s9

Step 2 — Split the cosine

A cosine is two exponentials, and each pairs with e^{-ikx} to give a clean Gaussian integral:

$$\cos k_0 x = \frac{1}{2}\left(e^{ik_0 x} + e^{-ik_0 x} \right)$$

$$A(k) = \frac{1}{2\sqrt{2\pi}} \int dx\; e^{-x^2/2L^2} \left[e^{-i(k-k_0)x} + e^{-i(k+k_0)x} \right]$$

Step 3 — Do the Gaussian integral

Standard result — the Fourier transform of a Gaussian is a Gaussian:

$$\int dx\; e^{-x^2/2L^2} e^{-iqx} = L\sqrt{2\pi}\, e^{-q^2L^2/2}$$

Apply it twice, with $q = k - k_0$ and $q = k + k_0$:

$$A(k) = \frac{L}{2}\left[e^{-(L^2/2)(k-k_0)^2} + e^{-(L^2/2)(k+k_0)^2} \right]$$

Two Gaussian humps, centred at $k = +k_0$ and $k = -k_0$, each of width $1/L$. The $+k_0$ hump becomes the right-moving pulse, the $-k_0$ hump the left-moving one.

Step 4 — Put A(k) into 7.90

$$u(x,t) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int dk\; A(k)\, e^{i[kx - \omega(k)t]} + \text{c.c.}$$ Lec 9, s3

With $\omega(k) = v(1 + a^2k^2/2)$ the exponent is **quadratic in k**, and A(k) is Gaussian — so the k-integral is Gaussian too and can be done exactly. The result is Jackson 7.98:

ANSWER — Jackson 7.98

$$u(x,t) = \frac{1}{2}\,\mathrm{Re}\left\{ \frac{\exp\left[-(x - va^2k_o t)^2 \,/\, 2L^2(1 + ia^2vt/L^2) \right]}{(1 + ia^2vt/L^2)^{1/2}}\, e^{i[k_o x - v(1+a^2k_o^2/2)t]} + (k_o \to -k_o) \right\}$$

Step 5 — Read the answer

Three things can be read straight off, without redoing any algebra:

① the pulse **centre** sits at $x = va^2k_o t$, so it travels at the **group velocity**

$$v_g = \frac{d\omega}{dk}\Big|_{k_o} = va^2k_o$$ Lec 8, s13

② the **carrier** $e^{i[k_o x - \omega(k_o)t]}$ moves at the phase velocity $\omega(k_o)/k_o$ — a different speed;

③ the **width** is no longer L but grows with time:

$$\text{width} = \sqrt{L^2 + (a^2vt/L)^2}$$ Lec 9, s6

A **short** pulse (small L) spreads fastest, because it needs a wide band of k and the different k travel at different speeds.